# 01 / POWER DISTRIBUTION

Supply: external regulated 24 V DC. Three independent switch-and-lamp circuits.

Blue = signal; red = +24 V; black = 0 V. 18 AWG simulated cuts; no current-rating claim.

Cockpit example: J1 pins 1-3 receive external switched +24 V; pins 4-6 return to 0 V.

Three toggle switches drive three 24 V indicators through one six-pin connector.

| | | | | | | |
|---|---|---|---|---|---|---|
| J1:1 (+24 V) | ——— | S1 MASTER | ——— | H1 indicator | ——— | J1:4 (0 V) |
| J1:2 (+24 V) | ——— | S2 PUMP | ——— | H2 indicator | ——— | J1:5 (0 V) |
| J1:3 (+24 V) | ——— | S3 LIGHTS | ——— | H3 indicator | ——— | J1:6 (0 V) |

For each branch, use its wire IDs on the I/O sheet and full endpoint schedule.

# 02 / 24 V DC INPUT AND OUTPUT WIRING

Three toggle switches and matching indicators. Each row identifies both connector pins and the cut wire.

| | | |
|---|---|---|
| J1:1 | W100 \| 18 AWG red \| .4 m | S1:1 |
| S1:2 | W101 \| 18 AWG red \| .4 m | H1:+ |
| H1:- | W102 \| 18 AWG black \| .8 m | J1:4 |
| J1:2 | W103 \| 18 AWG red \| .4 m | S2:1 |
| S2:2 | W104 \| 18 AWG red \| .4 m | H2:+ |
| H2:- | W105 \| 18 AWG black \| .8 m | J1:5 |
| J1:3 | W106 \| 18 AWG red \| .4 m | S3:1 |
| S3:2 | W107 \| 18 AWG red \| .4 m | H3:+ |
| H3:- | W108 \| 18 AWG black \| .8 m | J1:6 |

# 03 / TERMINAL AND WIRE SCHEDULE

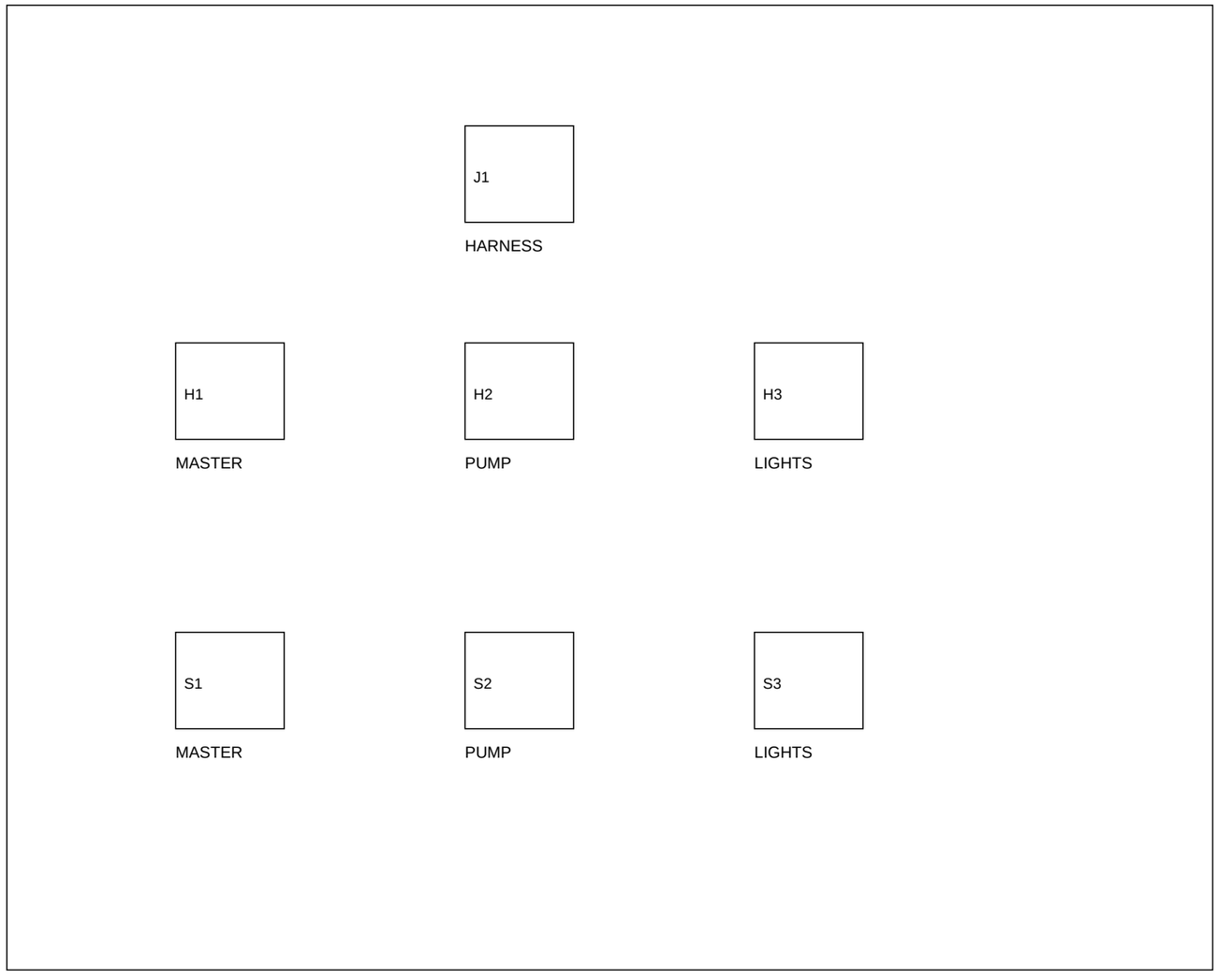
| PIN | WIRE | OTHER END | COLOUR | PIN | WIRE | OTHER END | COLOUR |
|---|---|---|---|---|---|---|---|
| J1:1 | 100 | S1:1 | red | | | | |
| S1:1 | 100 | J1:1 | red | | | | |
| S1:2 | 101 | H1:+ | red | | | | |
| H1:+ | 101 | S1:2 | red | | | | |
| H1:- | 102 | J1:4 | black | | | | |
| J1:4 | 102 | H1:- | black | | | | |
| J1:2 | 103 | S2:1 | red | | | | |
| S2:1 | 103 | J1:2 | red | | | | |
| S2:2 | 104 | H2:+ | red | | | | |
| H2:+ | 104 | S2:2 | red | | | | |
| H2:- | 105 | J1:5 | black | | | | |
| J1:5 | 105 | H2:- | black | | | | |
| J1:3 | 106 | S3:1 | red | | | | |
| S3:1 | 106 | J1:3 | red | | | | |
| S3:2 | 107 | H3:+ | red | | | | |
| H3:+ | 107 | S3:2 | red | | | | |
| H3:- | 108 | J1:6 | black | | | | |
| J1:6 | 108 | H3:- | black | | | | |

CSV and Excel schedules include PLC signal names, addresses and wire gauge.

# 04 / CONTROL PANEL LAYOUT

500 x 400 mm cockpit-style faceplate study. Switches below indicators; harness connector at the rear.

Coordinates and footprints come from devices.csv; dimensions are illustrative millimetres.




## PANEL REFERENCES

S1: SW-TOGGLE

S2: SW-TOGGLE

S3: SW-TOGGLE

H1: LED-24

H2: LED-24

H3: LED-24

J1: J-6

Reserve wiring ducts and clearances before any physical construction; no enclosure or heat study included.

# 05 / CONSOLIDATED BILL OF MATERIALS

Device quantities are consolidated by part. Wire totals are cut lengths, without spare/waste allowance.

| PART | DESCRIPTION | QTY | UNIT |
|---|---|---|---|
| J-6 | Six-pin panel connector | 1 | each |
| LED-24 | 24 V panel indicator | 3 | each |
| SW-TOGGLE | 24 V toggle switch | 3 | each |
| WIRE:18 AWG:black | 18 AWG black hookup wire | 2.4 | m |
| WIRE:18 AWG:red | 18 AWG red hookup wire | 2.4 | m |

Generic study parts are not manufacturer selections; PLC-24 represents the simulated controller, not a selected commercial model.